

Contents

This repository presents the SQL queries that were employed to extract insights related to **sales performance**, **regional sales trends**, and key metrics driving business growth for the **ADIDAS PRODUCT SALES ANALYSIS FOR THE YEAR 2020 AND 2021** project

Key SQL Skills:

- Joins
- Subqueries
- Window Functions
- Aggregation
- Group By

Explore the full project repository:

[ADIDAS PRODUCT SALES ANALYSIS FOR THE YEAR 2020 AND 2021](<https://github.com/yemifatodu/ADIDAS-PRODUCT-SALES-ANALYSIS-FOR-THE-YEAR-2020-AND-2021#adidas-product-sales-analysis-for-the-year-2020-and-2021>)

—

Use of Aliases:

I consider using shorter or more descriptive aliases for tables and columns. This approach makes my queries easier to read and manage, especially when I'm dealing with complex queries.

Indexing and Performance:

I make sure that the columns frequently used in WHERE, GROUP BY, and ORDER BY clauses (like 'invoice_date', 'Region', etc.) are indexed. This improves the performance of my queries, particularly when working with large datasets.

Partitioning Large Queries:

For Common Table Expressions (CTEs), I might consider partitioning the data by year and region. This can help with performance, especially when the dataset is large.

Handling NULL Values:

I ensure that my data doesn't contain NULL values in key columns like 'Total_Sales', 'Units_Sold', or 'Operating_Profit'. If there's a possibility, I use COALESCE to handle such cases.

Error Handling:

Depending on my SQL environment, I consider adding error-handling mechanisms or checks to ensure that my queries run smoothly without unexpected interruptions.

Additional Insights:

Using CTEs:

In this example, I use a CTE to calculate the total and average sales by region and year. This makes the query cleaner and potentially improves readability.

Combined Metrics:

For scenarios where I calculate multiple metrics (like 'TotalSales' and 'AverageSales') for the same grouping, I use a CTE to encapsulate the logic and reduce redundancy.

The queries I've prepared cover a broad range of analyses, from simple aggregations to more complex rankings and comparisons. The structure and explanations I've added make the SQL highly readable and easy to understand, which is essential for maintaining and sharing my work.

—

“sql

USE [YEMIFATODU2db];

```

-- Retrieve all columns from the ADIDDASSALE table
SELECT *
FROM [dbo].[ADIDDASSALE];
““

--

““sql
-- Calculate the overall total sales for both years combined
SELECT
SUM(Total_Sales) AS OverallTotalSales
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021);
““

--

““sql
-- Calculate the total sales for each region for the years 2020 and 2021
SELECT
YEAR(invoice__date) AS Year,
Region,
SUM(Total_Sales) AS TotalSales
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021)
YEAR(invoice__date),
Region
ORDER BY
Year,
Region;
““

--

““sql
-- Average Sales Per Region for 2020 and 2021
SELECT
Region,
AVG(Total_Sales) AS AverageSales
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021)
GROUP BY
Region;
““

--

““sql
-- Retrieve the total sales for each region, ordered by highest sales per year
SELECT
YEAR(invoice__date) AS Year,

```

```

Region,
SUM(Total_Sales) AS TotalSales
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021)
GROUP BY
YEAR(invoice__date),
Region
ORDER BY
Year,
TotalSales DESC;
““
—

““sql
-- Find the highest sales region for each year
SELECT
rs.Year,
rs.Region,
rs.TotalSales
FROM
(
SELECT
YEAR(invoice__date) AS Year,
Region,
SUM(Total_Sales) AS TotalSales
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021)
GROUP BY
YEAR(invoice__date),
Region
) rs
JOIN (
SELECT
Year,
MAX(TotalSales) AS MaxSales
FROM
(
SELECT
YEAR(invoice__date) AS Year,
Region,
SUM(Total_Sales) AS TotalSales
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021)
GROUP BY
YEAR(invoice__date),
Region
) subquery
GROUP BY

```

```

Year
) max_sales
ON rs.Year = max_sales.Year
AND rs.TotalSales = max_sales.MaxSales
ORDER BY
rs.Year;
““
--
““sql
-- Find the minimum sales in each region per year
;WITH RegionalSales AS (
SELECT
YEAR(invoice__date) AS Year,
Region,
SUM(Total__Sales) AS TotalSales
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021)
GROUP BY
YEAR(invoice__date),
Region
),
MinSales AS (
SELECT
Year,
MIN(TotalSales) AS MinSales
FROM
RegionalSales
GROUP BY
Year
)
SELECT
rs.Year,
rs.Region,
rs.TotalSales
FROM
RegionalSales rs
JOIN
MinSales ms
ON
rs.Year = ms.Year
AND rs.TotalSales = ms.MinSales
ORDER BY
rs.Year;
““
--
““sql
-- Average Sale in Each Region Per Year
SELECT
YEAR(invoice__date) AS Year,
Region,

```

```

AVG(Total_Sales) AS AverageSales
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice_date) IN (2020, 2021)
GROUP BY
YEAR(invoice_date),
Region
ORDER BY
Year,
Region;
““

--

““sql
-- Calculate Total and Average Sales for Each Region and Year
SELECT
YEAR(invoice_date) AS Year,
Region,
SUM(Total_Sales) AS TotalSales,
AVG(Total_Sales) AS AverageSales
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice_date) IN (2020, 2021)
GROUP BY
YEAR(invoice_date),
Region
ORDER BY
Year,
Region;
““

--

““sql
-- Total Units Sold for Each Year
SELECT
YEAR(invoice_date) AS Year,
SUM(Units_Sold) AS TotalUnitsSold
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice_date) IN (2020, 2021)
GROUP BY
YEAR(invoice_date)
ORDER BY
Year;
““

--

““sql
-- Total Units Sold for Each Region and Year
SELECT
YEAR(invoice_date) AS Year,

```

```

Region,
SUM(Units_Sold) AS TotalUnitsSold
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021)
GROUP BY
YEAR(invoice__date),
Region
ORDER BY
Year,
Region;
““

```

```

--
““sql
-- Calculate the Average Units Sold Per Region for Each Year
SELECT
YEAR(invoice__date) AS Year,
Region,
AVG(Units_Sold) AS AverageUnitsSold
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021)
GROUP BY
YEAR(invoice__date),
Region
ORDER BY
Year,
Region;
““

```

```

--
““SQL
-- The Region with the Highest Total Units Sold for Each Year (2020 and 2021)
;WITH RegionalUnits AS (
SELECT
YEAR(invoice__date) AS Year,
Region,
SUM(Units_Sold) AS TotalUnitsSold
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021)
GROUP BY
YEAR(invoice__date),
Region
),
RankedUnits AS (
SELECT
Year,
Region,
TotalUnitsSold,

```

```

ROW_NUMBER() OVER (PARTITION BY Year ORDER BY TotalUnitsSold DESC) AS Rank
FROM
RegionalUnits
)
SELECT
Year,
Region,
TotalUnitsSold
FROM
RankedUnits
WHERE
Rank = 1
ORDER BY
Year,
Region;
““

```

--

```

““SQL
-- The Region with the Lowest Total Units Sold for Each Year (2020 and 2021)
;WITH RegionalUnits AS (
SELECT
YEAR(invoice__date) AS Year,
Region,
SUM(Units__Sold) AS TotalUnitsSold
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021)
GROUP BY
YEAR(invoice__date),
Region
),
RankedUnits AS (
SELECT
Year,
Region,
TotalUnitsSold,
ROW_NUMBER() OVER (PARTITION BY Year ORDER BY TotalUnitsSold ASC) AS Rank
FROM
RegionalUnits
)
SELECT
Year,
Region,
TotalUnitsSold
FROM
RankedUnits
WHERE
Rank = 1
ORDER BY
Year;
““

```

--

```

““SQL
-- The Region with the Lowest Total Operating Profit for Each Year (2020 and 2021)
;WITH RegionalProfit AS (
SELECT
YEAR(invoice__date) AS Year,
Region,
SUM(Operating__Profit) AS TotalOperatingProfit
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021)
GROUP BY
YEAR(invoice__date),
Region
),
RankedProfit AS (
SELECT
Year,
Region,
TotalOperatingProfit,
ROW_NUMBER() OVER (PARTITION BY Year ORDER BY TotalOperatingProfit ASC) AS Rank
FROM
RegionalProfit
)
SELECT
Year,
Region,
TotalOperatingProfit
FROM
RankedProfit
WHERE
Rank = 1
ORDER BY
Year;
““

```

—

```

““SQL
-- Calculate the Total Operating Profit for Each Year
SELECT
YEAR(invoice__date) AS Year,
SUM(Operating__Profit) AS TotalOperatingProfit
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021)
GROUP BY
YEAR(invoice__date)
ORDER BY
Year;
““

```

—

```

““sql
-- Calculate the Average Operating Profit Per Region for Each Year

```

```

SELECT
YEAR(invoice__date) AS Year,
Region,
AVG(Operating__Profit) AS AverageOperatingProfit
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021)
GROUP BY
YEAR(invoice__date),
Region
ORDER BY
Year,
Region;
““

--
““sql
-- Regional Sales Breakdown for Each Year
;WITH RegionalData AS (
SELECT
YEAR(invoice__date) AS Year,
Region,
SUM(Total__Sales) AS TotalSales,
SUM(Units__Sold) AS TotalUnitsSold,
SUM(Operating__Profit) AS TotalOperatingProfit
FROM
[dbo].[ADIDDASSALE]
WHERE
Region IN ('South', 'West', 'Northeast', 'Southeast', 'Midwest')
AND YEAR(invoice__date) IN (2020, 2021)
GROUP BY
YEAR(invoice__date),
Region
),
OverallTotals AS (
SELECT
Year,
SUM(TotalSales) AS OverallTotalSales
FROM
RegionalData
GROUP BY
Year
)
SELECT
r.Year,
r.Region,
r.TotalSales,
r.TotalUnitsSold,
r.TotalOperatingProfit,
CAST(r.TotalSales AS FLOAT) / o.OverallTotalSales * 100 AS SalesPercentage
FROM
RegionalData r
JOIN
OverallTotals o

```

```

ON
r.Year = o.Year
ORDER BY
r.Year,
r.Region;
“‘
—

“‘sql
-- Regional Breakdown for Each Year (Example for South Region)
;WITH RegionalData AS (
SELECT
YEAR(invoice__date) AS Year,
Region,
SUM(Total__Sales) AS TotalSales,
SUM(Units__Sold) AS TotalUnitsSold,
SUM(Operating__Profit) AS TotalOperatingProfit
FROM
[dbo].[ADIDDASSALE]
WHERE
Region IN ('South')
AND YEAR(invoice__date) IN (2020, 2021)
GROUP BY
YEAR(invoice__date),
Region
)
“‘
—

“‘sql
-- Month-by-Month Sales Analysis for South Region
;WITH MonthlySales AS (
SELECT
YEAR(invoice__date) AS Year,
MONTH(invoice__date) AS Month,
SUM(Total__Sales) AS TotalSales
FROM
[dbo].[ADIDDASSALE]
WHERE
Region = 'South'
AND YEAR(invoice__date) IN (2020, 2021)
GROUP BY
YEAR(invoice__date),
MONTH(invoice__date)
),
HighestLowestSales AS (
SELECT
Year,
MAX(TotalSales) AS MaxSales,
MIN(TotalSales) AS MinSales
FROM
MonthlySales
GROUP BY
Year
)

```

```

SELECT
m.Year,
m.Month,
m.TotalSales,
CASE
WHEN m.TotalSales = h.MaxSales THEN 'Highest'
WHEN m.TotalSales = h.MinSales THEN 'Lowest'
ELSE 'In-between'
END AS SalesTrend
FROM
MonthlySales m
JOIN
HighestLowestSales h
ON
m.Year = h.Year
ORDER BY
m.Year, m.Month ASC;
--

--sql
-- Highest and Lowest Quarterly Sales for South Region
;WITH QuarterlySales AS (
SELECT
YEAR(invoice_date) AS Year,
DATEPART(QUARTER, invoice_date) AS Quarter,
SUM(Total_Sales) AS TotalSales
FROM
[dbo].[ADIDDASSALE]
WHERE
Region = 'South'
AND YEAR(invoice_date) IN (2020, 2021)
GROUP BY
YEAR(invoice_date),
DATEPART(QUARTER, invoice_date)
)
SELECT
Year,
Quarter,
MAX(TotalSales) AS HighestSales,
MIN(TotalSales) AS LowestSales
FROM
QuarterlySales
GROUP BY
Year,
Quarter
ORDER BY
Year, Quarter;
--

--sql
-- Highest and Lowest Quarterly Sales for West Region
WITH QuarterlySales AS (
SELECT

```

```

YEAR(invoice_date) AS Year,
DATEPART(QUARTER, invoice_date) AS Quarter,
SUM(Total_Sales) AS TotalSales
FROM
[dbo].[ADIDDASSALE]
WHERE
Region = 'West'
AND YEAR(invoice_date) IN (2020, 2021)
GROUP BY
YEAR(invoice_date),
DATEPART(QUARTER, invoice_date)
)
SELECT
Year,
Quarter,
MAX(TotalSales) AS HighestSales,
MIN(TotalSales) AS LowestSales
FROM
QuarterlySales
GROUP BY
Year,
Quarter
ORDER BY
Year, Quarter;
""

--
""sql
-- Highest and Lowest Quarterly Sales for Northeast Region
;WITH QuarterlySales AS (
SELECT
YEAR(invoice_date) AS Year,
DATEPART(QUARTER, invoice_date) AS Quarter,
SUM(Total_Sales) AS TotalSales
FROM
[dbo].[ADIDDASSALE]
WHERE
Region = 'Northeast'
AND YEAR(invoice_date) IN (2020, 2021)
GROUP BY
YEAR(invoice_date),
DATEPART(QUARTER, invoice_date)
)
SELECT
Year,
Quarter,
MAX(TotalSales) AS HighestSales,
MIN(TotalSales) AS LowestSales
FROM
QuarterlySales
GROUP BY
Year,
Quarter
ORDER BY

```

```

Year, Quarter;
““
--
““sql
-- Highest and Lowest Quarterly Sales for Southeast Region
;WITH QuarterlySales AS (
SELECT
YEAR(invoice_date) AS Year,
DATEPART(QUARTER, invoice_date) AS Quarter,
SUM(Total_Sales) AS TotalSales
FROM
[dbo].[ADIDDASSALE]
WHERE
Region = 'Southeast'
AND YEAR(invoice_date) IN (2020, 2021)
GROUP BY
YEAR(invoice_date),
DATEPART(QUARTER, invoice_date)
)
SELECT
Year,
Quarter,
MAX(TotalSales) AS HighestSales,
MIN(TotalSales) AS LowestSales
FROM
QuarterlySales
GROUP BY
Year,
Quarter
ORDER BY
Year, Quarter;
““
--
““sql
-- Highest and Lowest Quarterly Sales for Midwest Region
;WITH QuarterlySales AS (
SELECT
YEAR(invoice_date) AS Year,
DATEPART(QUARTER, invoice_date) AS Quarter,
SUM(Total_Sales) AS TotalSales
FROM
[dbo].[ADIDDASSALE]
WHERE
Region = 'Midwest'
AND YEAR(invoice_date) IN (2020, 2021)
GROUP BY
YEAR(invoice_date),
DATEPART(QUARTER, invoice_date)
)
SELECT
Year,
Quarter,

```

```

MAX(TotalSales) AS HighestSales,
MIN(TotalSales) AS LowestSales
FROM
QuarterlySales
GROUP BY
Year,
Quarter
ORDER BY
Year, Quarter;
““
—

““sql
-- Average Price per Unit of Products Sold for Each Year
SELECT
YEAR(invoice__date) AS Year,
AVG(Price__per__Unit) AS AveragePrice
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021)
GROUP BY
YEAR(invoice__date);
““
—

““sql
-- Minimum and Maximum Price per Unit of Products Sold for Each Year
SELECT
YEAR(invoice__date) AS Year,
MIN(Price__per__Unit) AS MinPrice,
MAX(Price__per__Unit) AS MaxPrice
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021)
GROUP BY
YEAR(invoice__date);
““
—

““sql
-- Correlation Between Price per Unit and Units Sold for Each Year
;WITH Stats AS (
SELECT
YEAR(invoice__date) AS Year,
[Price__per__Unit] AS PricePerUnit,
Units__Sold AS UnitsSold,
AVG([Price__per__Unit]) OVER (PARTITION BY YEAR(invoice__date)) AS AvgPrice,
AVG(Units__Sold) OVER (PARTITION BY YEAR(invoice__date)) AS AvgUnitsSold
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021)
),
Covariance AS (

```

```

SELECT
Year,
SUM((PricePerUnit - AvgPrice) * (UnitsSold - AvgUnitsSold)) AS CovXY,
SQRT(SUM((PricePerUnit - AvgPrice) * (PricePerUnit - AvgPrice))) AS StdDevX,
SQRT(SUM((UnitsSold - AvgUnitsSold) * (UnitsSold - AvgUnitsSold))) AS StdDevY
FROM
Stats
GROUP BY
Year
)
SELECT
Year,
CovXY / (StdDevX * StdDevY) AS CorrelationCoefficient
FROM
Covariance
ORDER BY
Year;
““
--
““sql
-- Correlation Between Price per Unit and Units Sold
SELECT
YEAR(invoice__date) AS Year,
CORR(Price__per__Unit, Units__Sold) AS CorrelationCoefficient
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021)
GROUP BY
YEAR(invoice__date);
““
--
““sql
-- Top Product Categories by Sales for Each Year
WITH CategorySales AS (
SELECT
YEAR(invoice__date) AS Year,
Product AS Category,
SUM(Total__Sales) AS SalesValue
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021)
GROUP BY
YEAR(invoice__date),
Product
)
SELECT
Year,
Category,
SalesValue
FROM
(

```

```

SELECT
Year,
Category,
SalesValue,
ROW_NUMBER() OVER (PARTITION BY Year ORDER BY SalesValue DESC) AS RowNum
FROM
CategorySales
) AS RankedCategories
WHERE
RowNum <= 3
ORDER BY
Year,
RowNum;
““

--
““sql
-- Top 3 Product Categories by Sales for the South Region
SELECT TOP 3
Product AS Category,
SUM(Total_Sales) AS SalesValue
FROM
[dbo].[ADIDDASSALE]
WHERE
Region = 'South'
GROUP BY
Product
ORDER BY
SalesValue DESC;
““

--
““sql
-- Top 3 Product Categories by Sales for the West Region
SELECT TOP 3
Product AS Category,
SUM(Total_Sales) AS SalesValue
FROM
[dbo].[ADIDDASSALE]
WHERE
Region = 'West'
GROUP BY
Product
ORDER BY
SalesValue DESC;
““

--
““sql
-- Top 3 Product Categories by Sales for the Northeast Region
SELECT TOP 3
Product AS Category,
SUM(Total_Sales) AS SalesValue
FROM
[dbo].[ADIDDASSALE]
WHERE

```

```

Region = 'Northeast'
GROUP BY
Product
ORDER BY
SalesValue DESC;
""

--
""sql
-- Top 3 Product Categories by Sales for the Southeast Region
SELECT TOP 3
Product AS Category,
SUM(Total_Sales) AS SalesValue
FROM
[dbo].[ADIDDASSALE]
WHERE
Region = 'Southeast'
GROUP BY
Product
ORDER BY
SalesValue DESC;
""

--
""sql
-- Top 3 Product Categories by Sales for the Midwest Region
SELECT TOP 3
Product AS Category,
SUM(Total_Sales) AS SalesValue
FROM
[dbo].[ADIDDASSALE]
WHERE
Region = 'Midwest'
GROUP BY
Product
ORDER BY
SalesValue DESC;
""

--
""sql
-- Top Product Categories by Profit for the West Region in 2020
SELECT TOP 3
Product,
SUM(Operating_Profit) AS TotalProfit
FROM
[dbo].[ADIDDASSALE]
WHERE
Region = 'West'
AND YEAR(invoice_date) = 2020
GROUP BY
Product
ORDER BY
TotalProfit DESC;
""

--

```

```

““sql
--Top Product Categories by Profit for the West Region in 2021
SELECT TOP 3
Product,
SUM(Operating_Profit) AS TotalProfit
FROM
[dbo].[ADIDDASSALE]
WHERE
Region = ‘West’
AND YEAR(invoice_date) = 2021
GROUP BY
Product
ORDER BY
TotalProfit DESC;
““

--

““sql
-- Top 3 Product Categories by Sales in South Region for 2020
;WITH RankedCategories AS (
SELECT
Product,
SUM(Total_Sales) AS TotalSales,
ROW_NUMBER() OVER (ORDER BY SUM(Total_Sales) DESC) AS Rank
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice_date) = 2020
AND Region = ‘South’
GROUP BY
Product
)
SELECT
Product,
TotalSales
FROM
RankedCategories
WHERE
Rank <= 3
ORDER BY
Rank;
““

--

““sql
-- Top 3 Product Categories by Sales in South Region for 2021
;WITH RankedCategories AS (
SELECT
Product,
SUM(Total_Sales) AS TotalSales,
ROW_NUMBER() OVER (ORDER BY SUM(Total_Sales) DESC) AS Rank
FROM
[dbo].[ADIDDASSALE]
WHERE

```

```

YEAR(invoice_date) = 2021
AND Region = 'South'
GROUP BY
Product
)
SELECT
Product,
TotalSales
FROM
RankedCategories
WHERE
Rank <= 3
ORDER BY
Rank;
““
—

““sql
-- Units Sold for Top 3 Product Categories in South Region for 2020
;WITH TopCategories AS (
SELECT
Product,
SUM(Total_Sales) AS TotalSales,
ROW_NUMBER() OVER (ORDER BY SUM(Total_Sales) DESC) AS Rank
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice_date) = 2020
AND Region = 'South'
GROUP BY
Product
),
CategoryUnits AS (
SELECT
Product,
SUM(Units_Sold) AS UnitsSold
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice_date) = 2020
AND Region = 'South'
GROUP BY
Product
)
SELECT
t.Product AS [Category],
c.UnitsSold AS [Units Sold]
FROM
TopCategories t
JOIN
CategoryUnits c
ON
t.Product = c.Product
WHERE

```

```

t.Rank <= 3
ORDER BY
t.Rank;
““
—

““sql
-- Units Sold for Top 3 Product Categories in South Region for 2021
;WITH TopCategories AS (
SELECT
Product,
SUM(Total_Sales) AS TotalSales,
ROW_NUMBER() OVER (ORDER BY SUM(Total_Sales) DESC) AS Rank
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice_date) = 2021
AND Region = 'South'
GROUP BY
Product
),
CategoryUnits AS (
SELECT
Product,
SUM(Units_Sold) AS UnitsSold
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice_date) = 2021
AND Region = 'South'
GROUP BY
Product
)
SELECT
t.Product AS [Category],
c.UnitsSold AS [Units Sold]
FROM
TopCategories t
JOIN
CategoryUnits c
ON
t.Product = c.Product
WHERE
t.Rank <= 3
ORDER BY
t.Rank;
““
—

““sql
-- Total Sales by Retailer for 2020 and 2021
SELECT
Retailer,
YEAR(invoice_date) AS Year,

```

```

SUM(Total_Sales) AS TotalSales
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021)
GROUP BY
Retailer,
YEAR(invoice__date)
ORDER BY
Retailer,
Year;
““

```

—

```

““sql
-- Sales Comparison for the Top 5 Retailers Between 2020 and 2021
;WITH RetailerSales AS (
SELECT
Retailer,
YEAR(invoice__date) AS Year,
SUM(Total_Sales) AS TotalSales
FROM
[dbo].[ADIDDASSALE]
WHERE
YEAR(invoice__date) IN (2020, 2021)
GROUP BY
Retailer,
YEAR(invoice__date)
),
TopRetailers AS (
SELECT
Retailer,
SUM(TotalSales) AS TotalSales
FROM
RetailerSales
GROUP BY
Retailer
ORDER BY
TotalSales DESC
OFFSET 0 ROWS
FETCH NEXT 5 ROWS ONLY
)
SELECT
r.Retailer,
r.Year,
r.TotalSales
FROM
RetailerSales r
JOIN
TopRetailers t
ON
r.Retailer = t.Retailer
ORDER BY
r.Retailer,

```

r.Year;
“